UTM UNIVERSITI TEKNOLOGI MALAYSIA | Faculty of Computing

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| Course Code | : | **SCST1223** |
| Course Name | : | **PROBABILITY AND STATISTICAL DATA ANALYSIS** |
| Lecturer | : | SHAMINI A/P RAJA KUMARAN |
| Academic Session | : | 2025-2026/2 |
| Assessment Title | : | Homework Week 1 (Introduction to Statistics) |
| Submission Due Date | : | Wednesday, 25 March 2026, 5:00 PM |

Prepared by :

| Matric No. | Student Name |
|---|---|
| A23CS4026 | MOHAMMED MOHSEN MOHAMMED ALSAKKAF |

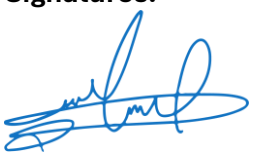
Assessment Mode : Individual

| Feedback from Lecturer : |
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**Student Declaration**

I hereby declare that this assessment is my own original work. I have not plagiarised any sources and have not used generative AI tools unless explicitly permitted by the lecturer. All references used have been properly acknowledged.

**Signatures:**



# Question 1

Choose from: **Nominal / Ordinal / Interval / Ratio**. Identify the **type of data** for each of the following:

a) **Student ID number :** -> Answer = **Nominal** because it is numerical label used for identification only and the numbers have no mathematical value

b) **Ranking in a competition (1st, 2nd, 3rd)** -> Answer = **Ordinal** because the data has specific order or ranking

c) **Temperature of a city in °C** -> Answer = **Interval** because the data is ordered with equal intervals, but the 0 does not represent a true zero

d) **Gender (Male/Female)** -> Answer = **Nominal** because the categories with no inherent mathematical order so it's Nominal

# Question 2

Determine whether the following represent **Qualitative or Quantitative data**.

a) **Blood type of a patient** -> Answer = **Qualitative** because this deals with descriptions or such like categories

b) **Number of books owned by a student** -> Answer = **Quantitative** because this deals with the numerical counts

c) **Colour of a car** -> Answer = **Qualitative** because the colors deals with the description or the attributes

d) **Height of a person** -> Answer = **Qualitative** because this deals with the numerical measurement

## Question 3

Classify the following as **Discrete or Continuous data**.

a) **Number of students in a class** -> Answer = **Discrete** because the values are countable whole numbers

b) **Weight of a person** -> Answer = **Continuous** because the values can take any point within a range such including decimals

c) **Number of cars in a parking lot** -> Answer = **Discrete** because the Number of the cars are countable individual units

d) **Time taken to complete an exam** -> Answer = **Continuous** because this Time is a measurement that can be broken down into infinite fractions of seconds

## Question 4

Identify whether the data represents **Population or Sample**.

a) **Survey of all students in UTM** -> Answer = **Population** because this includes every member of the group being studied

b) **Survey of 50 students from UTM** -> Answer = **Sample** because this is a subset or small part of the larger group

c) **Measurement of all employees in a company** -> Answer = **Population** because this includes the entire collection of employees

d) **Study of 30 customers from a supermarket** -> Answer = **Sample** because this is a small group selected to represent the total customers base

**THE END**